

Filesystems & Denny's + LQ scraping

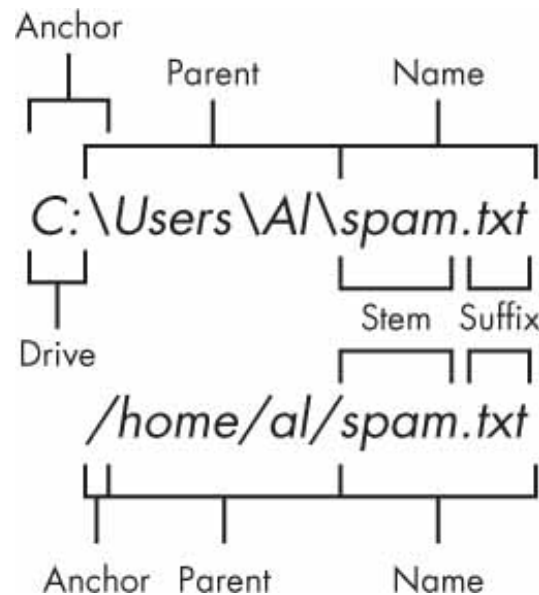
Lecture 14

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Filesystems

Pretty much all commonly used operating systems make use of a hierarchically structured filesystem.

This paradigm consists of directories which can contain files and other directories (which can then contain other files and directories and so on).



Absolute vs relative paths

Paths can either be absolute or relative, and the difference is very important. For portability reasons you should almost never use absolute paths.

Absolute path examples

- 1 `/var/ftp/pub`
- 2 `/etc/samba.smb.conf`
- 3 `/boot/grub/grub.conf`

Relative path examples

- 1 `Sta523/filesystem/`
- 2 `data/access.log`
- 3 `filesystem/nelle/pizza.cfg`

Special directories

```
1 dir(path = "data/")
```

```
[1] "ak"                "gis"                "lego_sales.rds"
[4] "movies"            "office_ratings.csv" "phone.csv"
[7] "pvec_res.Rdata"    "us"
```

```
1 dir(path = "data/", all.files = TRUE)
```

```
[1] "."                ".."                 ".DS_Store"
[4] "ak"               "gis"               "lego_sales.rds"
[7] "movies"           "office_ratings.csv" "phone.csv"
[10] "pvec_res.Rdata"   "us"
```

```
1 dir(path = "../")
```

```
[1] "css"    "slides"
```

```
1 dir(path = "data/../../")
```

```
[1] "css"    "slides"
```

```
1 dir(path = "../../")
```

```
[1] "CLAUDE.md"      "config.yaml"    "data"           "docs"
[5] "layouts"        "Makefile"       "README.md"      "resources"
[9] "static"         "website.Rproj"
```

Home directory and ~

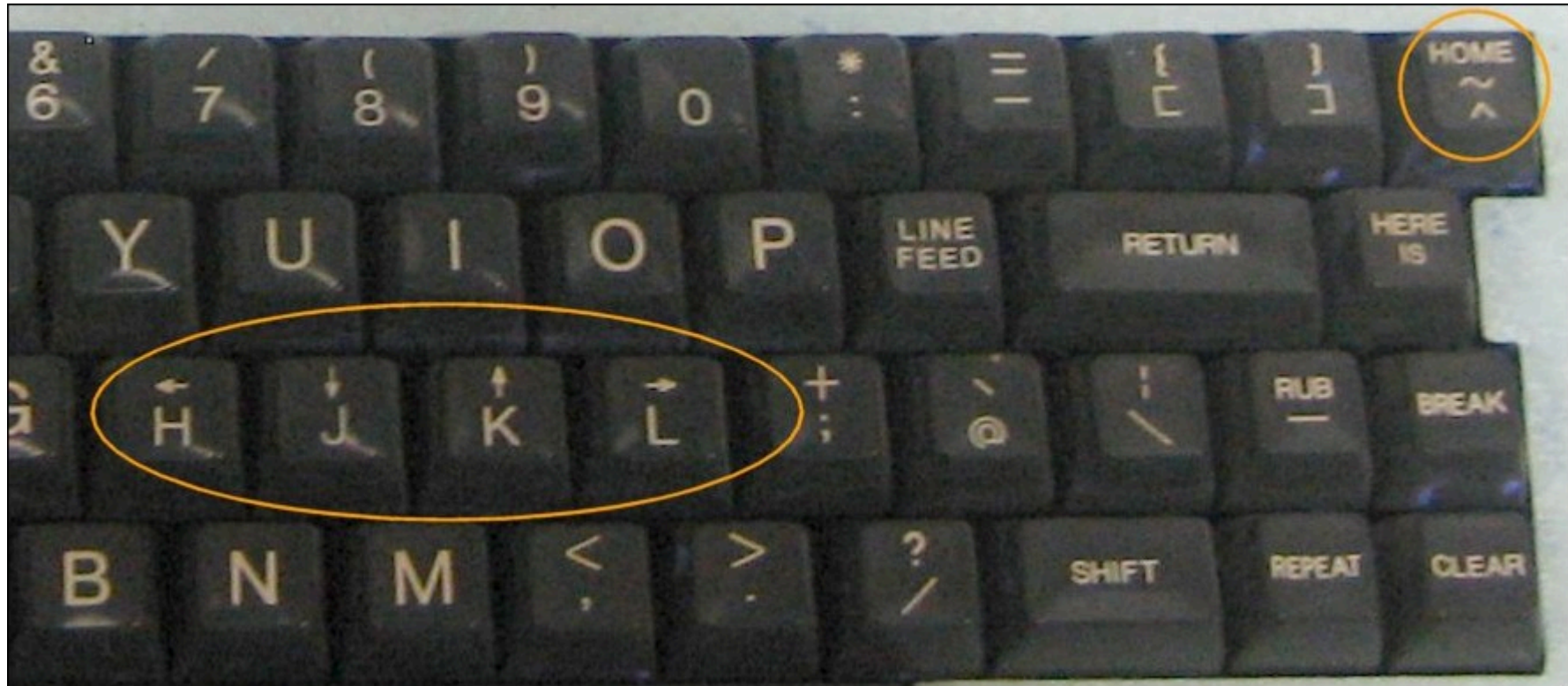
Tilde (~) is a shortcut that expands to the name of your home directory on unix-like systems.

```
1 dir(path = "~/Desktop/Sta523-Sp25/website/")
```

character(0)

Why ~?

Below is the keyboard from an [ADM-3A](#) terminal from the 1970s,



Working directories

R (and OSes) have the concept of a working directory, this is the directory where a program / script is being executed and determines the absolute path of any relative paths used.

```
1 getwd()
```

```
[1] "/Users/rundel/Desktop/Sta323-Sp26/website/static/slides"
```

```
1 setwd("~/")  
2 getwd()
```

```
[1] "/Users/rundel"
```


If the first line of your R script is

```
setwd("C:\\Users\\jenny\\path\\that\\only\\I\\have")
```

I* will come into your office and
SET YOUR COMPUTER ON FIRE 🔥.

* or maybe Timothée Poisot will

RStudio and Working Directories

Just like R, RStudio also makes use of a working directory for each of your sessions - we haven't had to discuss these yet because when you use an RStudio project, the working directory is automatically set to the directory containing the [Rproj](#) file.

This makes your project portable as all you need to do is to send the project folder to a collaborator (or push to GitHub) and they can open the project file and have identical *relative* path structure.

Some useful base R filesystem functions

- `dir()` - list the contents of a directory
- `basename()` - Removes all of the path up to and include the last path separator (/)
- `dirname()` - Returns the path up to but excluding the last path separator
- `file.path()` - a useful alternative to `paste0()` when combining paths (and urls) as it will add a / when necessary.
- `unlink()` - delete files and or directories
- `dir.create()` - create directories
- `fs` package - collection of filesystem related tools based on unix cli tools (e.g. `ls`)



The fs package

The fs package provides a cross-platform, uniform interface to file system operations. It wraps system calls with consistent naming and behavior across operating systems.

```
1 library(fs)
```

Key features:

- All functions are vectorized
- Provides consistent & helpful error messages on failure
- Handles paths consistently across Windows, Mac, and Linux
- Consistent naming scheme via function prefixes (`file_`, `dir_`, `path_`)

Path manipulation - `path()`

The `path()` function constructs file paths “correctly” for the current OS:

```
1 path("data", "myfile.csv")
```

data/myfile.csv

```
1 path("~", "Desktop/", "/Sta523-Fa25")
```

~/Desktop/Sta523-Fa25

Other useful path functions:

- `path_abs()` - convert to absolute path
- `path_expand()` - expand `~` in paths
- `path_file()` - extract filename (like `basename()`)
- `path_dir()` - extract directory (like `dirname()`)
- `path_ext()` - extract file extension
- `path_ext_set()` - change file extension

Listing files and directories

`dir_ls()` lists directory contents with more information than `dir()`:

```
1 dir_ls("data/")
```

```
data/ak          data/gis          data/lego_sales.rds
data/movies      data/office_ratings.csv data/phone.csv
data/pvec_res.Rdata data/us
```

file lists can also be filtered via glob, regex, or type:

```
1 dir_ls("data/", glob = "*.csv")
```

```
data/office_ratings.csv data/phone.csv
```

```
1 dir_ls("data/", type = "file", recurse = TRUE)
```

```
data/ak/states.dbf
data/ak/states.prj
data/ak/states.shp
data/ak/states.shx
data/gis/AnneArundel/AnneArundelN.dbf
data/gis/AnneArundel/AnneArundelN.prj
data/gis/AnneArundel/AnneArundelN.qpj
data/gis/AnneArundel/AnneArundelN.shp
data/gis/AnneArundel/AnneArundelN.shx
data/gis/AnneArundel/AnneArundelN84.dbf
data/gis/AnneArundel/AnneArundelN84.prj
data/gis/AnneArundel/AnneArundelN84.qpj
```

data/gis/AnneArundel/AnneArundelN84.shp
data/gis/AnneArundel/AnneArundelN84.shx
data/gis/airports/airports.dbf
data/gis/airports/airports.prj
data/gis/airports/airports.sbn
data/gis/airports/airports.sbx
data/gis/airports/airports.shp
data/gis/airports/airports.shx
data/gis/airports/airports.txt
data/gis/airports/airports.xml
data/gis/airports/airports.dbf

Creating and deleting

Creating directories:

```
1 dir_create("data/temp")  
2 dir_create("data/temp/a/b/c")
```

Deleting files and directories:

```
1 file_delete("data/myfile.csv")  
2 dir_delete("data/temp")
```

Copying and moving

Copy files or directories:

```
1 file_copy("data/file.csv", "backup/file.csv")
2 dir_copy("data/", "backup/")
```

Move (rename) files or directories:

```
1 file_move("data/old.csv", "data/new.csv")
```

Check if files/directories exist:

```
1 file_exists("data/myfile.csv")
2 dir_exists("data/temp")
```

File metadata

Get information about files:

```
1 file_info("data/office_ratings.csv")
```

```
# A tibble: 1 × 18
```

	path	type	size	permissions	modification_time	user	group	device_id
	<fs::path>	<fct>	<fs::>	<fs::perms>	<dtm>	<chr>	<chr>	<dbl>
1	...ratings.csv	file	7.02K	rw-r--r--	2022-08-29 10:47:54	rund...	staff	16777229

```
# i 10 more variables: hard_links <dbl>, special_device_id <dbl>, inode <dbl>,  
#   block_size <dbl>, blocks <dbl>, flags <int>, generation <dbl>,  
#   access_time <dtm>, change_time <dtm>, birth_time <dtm>
```

We can use vectorization to get info on multiple files,

```
1 dir_ls("data/") |>  
2   file_info()
```

```
# A tibble: 8 × 18
```

	path	type	size	permissions	modification_time	user	group	device_id
	<fs::path>	<fct>	<fs::b>	<fs::perms>	<dtm>	<chr>	<chr>	<dbl>
1	data/ak	dire...	192	rwxr-xr-x	2023-10-11 22:08:38	rund...	staff	16777229
2	data/gis	dire...	256	rwxr-xr-x	2024-11-13 09:58:14	rund...	staff	16777229
3	...sales.rds	file	23.07K	rw-r--r--	2022-08-29 10:47:54	rund...	staff	16777229
4	...ta/movies	dire...	128	rwxr-xr-x	2023-10-11 22:08:38	rund...	staff	16777229
5	...tings.csv	file	7.02K	rw-r--r--	2022-08-29 10:47:54	rund...	staff	16777229
6	...phone.csv	file	99	rw-r--r--	2022-08-29 10:47:54	rund...	staff	16777229
7	...res.Rdata	file	369	rw-r--r--	2022-08-29 10:47:54	rund...	staff	16777229
8	data/us	dire...	320	rwxr-xr-x	2023-10-11 22:08:38	rund...	staff	16777229

```
# i 10 more variables: hard_links <dbl>, special_device_id <dbl>, inode <dbl>,  
# block_size <dbl>, blocks <dbl>, flags <int>, generation <dbl>,  
# access_time <dtm>, change_time <dtm>, birth_time <dtm>
```

Denny's and LQ Scraping Demo